

Entropy-based Planning Algorithm for Personalized Interventions: Preliminary Report on Question-reduction Agent

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ABSTRACT

Prevention of lifestyle diseases is an important issue in terms of both the lifelong well-being of each individual and the burden on society. Although several artificial-intelligence-based chatbots have been proposed to support preventing such diseases, they have not yet reached a complex problem setting that combines multiple theories of behavioral change. In such complex settings, in addition to intervention selection, it is important to reduce the amount of attributes acquired to reduce the user's burden in preventing inhibition of motivation. Therefore, we developed the Entropy-Based Planning algorithm that selects interventions from a small number of questions. To evaluate the algorithm, we conducted a short-term randomized control trial (RCT) with a small number of users as an initial validation. Due to certain limitations, we were unable to obtain results indicating sufficient effectiveness. However, important findings were obtained for the future development of behavioral-change support bots. The findings include the need to investigate the relevance of preferences among multiple behavior-change theories and balance the number of questions and accuracy of personalization on user impressions. We intend to conduct further research on the basis of these findings as well as conduct a large-scale RCT after improving the algorithm and experiment settings.

CCS CONCEPTS

• **Applied computing** → **Health care information systems**;
Health care information systems.

KEYWORDS

Healthcare, Behavior Change, User Burden, Planning Algorithm

ACM Reference Format:

Taiga Sano, Masami Takahashi, Kaori Fujimura, Takafumi Inoue, Tae Sato, and Yurika Katagiri. 2024. Entropy-based Planning Algorithm for Personalized Interventions: Preliminary Report on Question-reduction Agent. In *Companion of the 2024 ACM International Joint Conference on Pervasive*

and Ubiquitous Computing Pervasive and Ubiquitous Computing (UbiComp Companion '24), October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3677590>

1 INTRODUCTION

Across many nations, the imperative of mitigating lifestyle diseases, encompassing conditions such as cancer, cardiovascular disease, cerebrovascular incidents, and diabetes, has emerged as an area of escalated societal concern. These maladies greatly contribute to the worldwide mortality rate and profoundly influence the patient's life-long well-being and quality of life. They will also have a significant impact on the increase in healthcare costs borne by the government. Therefore, it is of critical importance to stimulate lifestyle enhancements among those susceptible to these diseases. However, fostering tailored behavior alterations congruent with individual attributes remains formidable.

Several artificial intelligence (AI) chatbots have been proposed to personalize interventions [17, 21]. Despite the remarkable progress in AI technology, many chatbots still take scenario-based strategies. Such strategies may have a limited range of user attributes and only provide simplified personalization according to a specific behavioral-change theory. Motivations through dialogue conducted by humans are a more complex problem that combines multiple theories and requires more attribute items to be considered [3, 7]. However, combining multiple theories increases the number of attributes that users are asked about and may increase user burden. This can be the factor that inhibits motivation[2]. We hypothesize that question reduction of the necessary information from a wide range of candidate attributes would be useful for solving such a problem without increasing the burden on the user.

Following the guidelines of human-computer interaction (HCI) research for evaluating behavioral-change support systems[10], we conducted an initial experiment with a very small number of users and developed a simple planning algorithm for question reduction and intervention selection, the Entropy-Based Planning (EBP) algorithm, as an evaluation target. The results indicate that the design of personalized message categories could be improved for large-scale experiments. We also found that the balance between the reduction in the number of questions and accuracy of personalization may worsen the user's impression. This is something that should be fully verified in the future and will be an important finding regarding

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UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia

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ACM ISBN 979-8-4007-1058-2/24/10

<https://doi.org/10.1145/3675094.3677590>

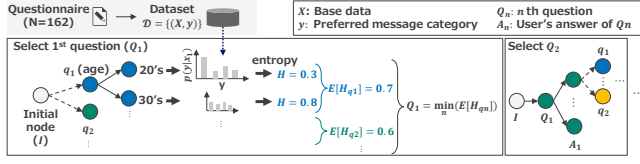


Figure 1: Overview of EBP algorithm

the design and experimentation of personalized AI for behavioral change.

2 RELATED WORKS

Supporting lifestyle change with AI chatbots is expected to be an alternative to human experts [17, 21]. Most studies used scenario-based strategies built according to specific motivational theories [4, 18]. There are also applications that use reinforcement learning to optimize messages and other information [1, 9]. These studies selected optimal interventions on the basis of attributes such as the user's health status, self-efficacy, and knowledge about the disease as states of reinforcement learning agents. Our study differs from these studies in that we assume a broader range of user attributes and incorporates the problem of extracting some of these attributes through queries.

3 METHOD

3.1 Design of Questions and Messages

We developed an AI agent that ask for attributes to predict the category of intervention message that the user most prefers, starting from a point where the AI agent has no information about the user. We used 32 attributes, e.g., age, personality[16], and health knowledge. This is a larger number of items than defined for most current AI chatbots. The messages were designed to suppress time discounting [8], considering the impact it has on habits. Nineteen exercise promotion messages were prepared that were, associated with four categories: increase the value of the exercise (increase exercise: IE), reduce the value of not performing the exercise (reduce no exercise: RE), increase the future health value (increase health: IH), and reduce the discounted future health value (reduce discount health: RH). Each message may be phrased in a way that is thought to enhance its content (e.g., using a gain frame to explain the value of a recommended behavior). Table 1 lists examples of messages of each category. Our ultimate goal is to develop an effective intervention selection algorithm combining multiple behavioral-change theories. However, as an initial test, we focused on selecting which elements of the time-discount suppression should be focused on in the intervention. The exercise to be promoted was at an exercise intensity of 3.5 to 4 metabolic equivalents.

3.2 Entropy-Based Planning Algorithm

The proposed EBP algorithm for AI agents involves optimizing the number of questions. The goal of the agent is to increase the confidence in the user's intervention preference with fewer questions and maximize motivation after the intervention. The agent acquires policies from data $\mathcal{D} = \{(X, y)\}$ generated from the questionnaire (Fig. 1). The questionnaire data are obtained in

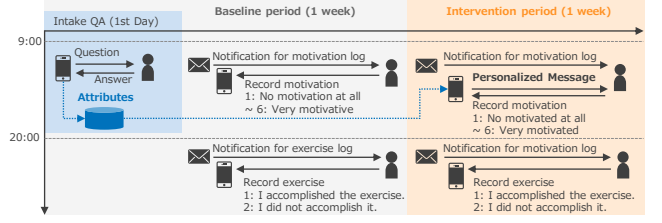


Figure 2: Experimental Flow: Motivation and exercise logs were recorded during the baseline and intervention periods.

advance, and there are 162 records. Each record contains the answers $X = (x_1, x_2, \dots, x_{32})$ to 32 questions $q_1, q_2, \dots, q_{32}$, where y indicate the intervention that the person rated as most preferred. The questionnaire is based on a 6-point Likert scale on whether the user would be willing to perform the exercise that they said they would be most likely to perform when presented with each message. The values were averaged for each message category, and the category with the highest average value was designated as y .

The agent constructs a policy as a tree, the nodes of which are the question Q and the user's answer A . The top of the tree is the initial node (I). When selecting Q_n , questions that have not been selected by Q_{n-1} are considered as candidates. For example, as shown in Fig 1, all questions are candidates since none have been asked when Q_1 is selected. For all the candidate questions, we create groups for which each option is selected in the questionnaire and find the probability distribution of the occurrence of each y in each group. For example, if q_1 is an age-related question and there are options for each age group, we obtain $p(y|x_1 = 20)$ for a group of respondents whose x_1 is "20s" and $p(y|x_1 = 30)$ for a group of "30s". For each group, the entropy H of the obtained probability distribution is calculated and averaged over all choices to obtain $E[H_{qn}]$. The question with the lowest $E[H_{qn}]$ (i.e., the question that shifts the distribution of the preference away from a uniform distribution) is chosen as Q_1 .

The same procedure is used to select Q_2 with each node of choice A_1 of Q_1 as a parent node. Note that only the record that selects each option is subject to the subsequent entropy calculation. This process is repeated until the entropy becomes zero and the intervention is uniquely determined, or until there are no more data corresponding to the condition. The intervention with the highest probability at the end of the process is implemented as the optimal intervention. We trained an agent that selects an intervention in at most eight questions using this algorithm.

3.3 Experimental Protocols

We developed a web-application system that asks questions to the user and presents intervention messages following the EBP algorithm. With this system, we evaluated whether the EBP algorithm promotes user motivation to exercise and actual exercise implementation. The target exercise was selected by each user in the questionnaire, but "a fast walk for 15 minutes" was chosen to unify conditions such as the effect of weather conditions. The duration of the exercise was also changed from 30 to 15 minutes in consideration of the difficulty level of the actual action.

Table 1: Message categories and example messages

Category of message	Example message
Increase Exercise(IE)	30 minutes of light exercise will burn about 100 kcal.
Reduce no Exercise (RE)	If you are watching TV, you can only consume about 26 kcal in 30 minutes.
Increase Health (IH)	Imagine 10 years from now. If you are healthy, you will be able to continue doing the things that you enjoy now (e.g., drinking, sports, etc.).
Reduce discount Health (RH)	You seem to be in good health today. If your health continues to be good, you will be able to have fun.

The flow of the experiment is shown in Fig. 2. The experiment was divided into a baseline period without messages and an intervention period with messages. At the beginning of the experiment, there was an interactive intake question answering in which the user answered a question from the system. This question was optimized according to the answer in accordance with the EBP algorithm. Therefore, the system dynamically selected optimal questions on the basis of the user’s answers as chatbots do. During the baseline period, users answered the motivation log every morning and the exercise log every evening. The motivation log was on a 6-point Likert scale, and the exercise log was a binary of performed or not performed. During the intervention period, users recorded their motivation after they had seen the intervention message and responded to the exercise log that evening. The messages sent during the intervention period were seven messages randomly selected from the message category chosen with the EBP algorithm. Since each category had 2–4 messages, duplicate messages could be presented.

4 EXPERIMENT

4.1 Conditions

We tested whether an agent with a small number of questions can present messages that reflect individual preferences better than when messages are presented randomly. The experiment was conducted as a randomized controlled trial (RCT). The participants were HCI researchers who were in good health and able to exercise. By including such researchers, we also expected to receive feedback on the experimental design. Since the system was displayed in Japanese, the participants were required to be Japanese speakers.

We recruited ten participants who were not currently receiving treatment for lifestyle diseases from a physician. We then divided the participants into two groups: the Entropy group, in which messages were personalized and the number of questions was minimized using the EBP algorithm, and Random group (control group). In the Random group, one message category was randomly selected and messages within that category was presented. Because the message category was randomly selected, personalization questions were not necessary for Random group, but all 32 questions were asked to investigate the effect of fewer questions. We excluded one participant from the Random group because the participant was dually enrolled in different email addresses and assigned to the Entropy group and the Random group, respectively. Participants who lacked either motivation or behavior logs for a majority of the days (i.e., four or more days) at the end of the two-week experimental period were considered dropouts and excluded. This resulted in the exclusion of one participant from the Random group. Finally,

Table 2: Changes in average motivation by period

Group	Ave. motivation in each period (std.)	
	Baseline	Intervention
Entropy	3.7 (0.58)	4.0 (0.68)
Random	3.7 (0.87)	3.9 (0.53)

Table 3: Changes in exercise days rate by period

Group	Exercise days in each period (std.)	
	Baseline	Intervention
Entropy	5.0 (1.58)	4.6 (2.88)
Random	3.7 (0.58)	3.3 (0.52)

a total of eight participants, five in the Entropy group and three in the Random group, were eligible for evaluation.

After the experiment, semi-structured interviews were conducted with participants who gave permission to be interviewed about the content of the messages and their motivation during each period. Interviews were conducted with a total of five participants, three in the Entropy group and two in the Random group. The interviews were analyzed by four researchers in the following procedure with reference to the affinity diagram[6, 13] to obtain hypotheses for improving the question–reduction agent. First, factors that promoted and inhibited exercise were extracted from the transcripts. Next, the extracted statements were grouped and labeled, and the factors that promoted or inhibited the behavior were discussed. Extraction from the transcripts was conducted independently by individual researchers, while labeling and grouping were conducted jointly.

4.2 Results

A comparison of the motivation logs by period and group is shown in Table 2. The values of the motivation logs for each participant for one week were averaged, and the mean and standard deviation (std.) were taken for each period and group. The Entropy group showed a slightly larger increase in motivation during the intervention period. The effect size of using the EBP algorithm was measured using difference in difference (DiD) and was 0.10. Therefore, we conclude that the EBP algorithm may increase motivation, but only by a small amount.

A comparison of the exercise logs by period and group is shown in Table 3. The percentage of days in each participant’s exercise log in which exercises were performed during the week was calculated,

and the mean and std. were taken for each period and group. Days, when no exercise log was recorded, were considered those in which no exercise was performed. The percentage of exercises performed decreased during the intervention period in both groups. The effect size of the EBP algorithm obtained from DiD was -0.01 .

5 DISCUSSION

The results of the experiment belied our hypothesis, especially in the implementation of behavior. We discuss what kept participants from exercising. Note that the interviews were conducted with a small number of participants for the purpose of obtaining hypotheses. We plan to verify the validity of these hypotheses and ensure their validity.

5.1 Limitations

The influence of external factors may be the reason for the drop in behavioral performance. In the post-experiment questionnaire, two participants from the Entropy group mentioned that they did not exercise because of their illness or an infection in the family. In the COM-B model [20], which models the process of behavioral change, opportunity is considered to be a requirement to perform the behavior along with motivation. This is also consistent with the discrepancy between the increase in the motivation log and decrease in the action log. In the interviews, several participants mentioned the influence of goal setting and recording on motivation. It is known that these factors influence behavior [11, 12]. It is possible that boredom with these elements may have had a more negative effect on the promotion of exercise than the motivational effect of the messages in the short two-week experiment. To conduct large-scale RCTs, it is necessary to cover a wider population and ensure that the number of participants and length of the experiment are large enough to reduce the influence of external factors.

5.2 Inadequacy of Message Category Separation

It is possible that the messages were not adequately categorized because the category was based solely on the suppression of time-discounting since this was an initial validation. According to the results of the interview with a participant in the Entropy group (ID9), messages that showed specific actions during the exercise (“Pulling the elbows backward significantly during exercise uses the pectoral muscles and helps prevent shoulder stiffness.”) were motivating (“*I was motivated by specific exercise instructions*”), but messages with negative wording (“If you do not exercise for 15 minutes you will not consume 50 kcal.”) made the subject feel coerced and had a negative impression (“*I felt compelled by the negative message*”). Both messages belong to IE but are expressed differently to improve their value in terms of framing [5]. Similarly, a participant in the Random group (ID6) perceived the specific message (“If you are still healthy 10 years from now, you can continue to do the things you enjoy now in your daily life.”) positively (“*I was motivated because health 10 years from now was a new perspective.*”) but the vague message (“If your body continues to be in good health, you are going to have a lot of fun.”) negatively (“*I didn’t feel anything, not much impression.*”), although both of them belongs to RH.

Viewpoints different from the theory of time discounting, such as preferences for gain or loss-framed messages and preferences in the specificity of content, also affected the evaluation of messages. These were introduced to emphasize the content of the text, but it seems that the user characteristics also influenced the preference for them. This is consistent with our ultimate goal of developing an optimal intervention-selection system from a combination of multiple behavioral-change techniques (BCTs). We need to investigate the preference relationships among multiple BCT categories (e.g., defined in the CALO-RE taxonomy [15]) and design personalized message lists that take these relationships into account.

5.3 Risk of Question Reduction

There were also findings that were obtained due to inadequate personalization. Excessively reducing the number of questions may worsen the user’s impression of the personalization accuracy when personalization is inadequate. In this study, we expected to reduce user burden by reducing the number of questions. The EBP algorithm succeeded in reducing the number of questions in the Entropy group (four questions on average). However, participant ID9, who was asked very few questions (two questions), expressed mistrust of the small number of questions asked in the interview (“I thought it was too few questions.”). We wonder if this negative effect is solely due to the small number of questions. As mentioned above, this participant also had a negative impression of the message. It is possible that the poor personalization accuracy was perceived to be due to the small number of questions, which may have worsened the impression. In a movie recommendation system, user burden was allowed when the quality of recommendations was sufficient [14]. Others have reported that a certain degree of input burden contributes to habit formation in applications that support self-monitoring [19]. We need to further investigate the relationship among intervention-personalization accuracy, number of questions asked during intake (i.e., user burden), and user’s impression of them, in healthcare message settings.

6 CONCLUSION

We conducted a small RCT to examine the experimental design for testing a system to personalize an intervention from a small number of questions. We proposed the EBP algorithm and presented personalized exercise-promotion messages. While we could not confirm the efficacy of the algorithm within the scope of this work, the need for personalization that combines multiple behavioral-change theories and the importance of investigating the relationship among such theories, as suggested by our interview results, is a critical problem for the research community on behavior-change support bots to study in the future. Future research will be conducted in line with these findings. We also aim to develop and determine the effectiveness of an improved question reduction algorithm, when personalization is sufficiently implemented by modifying the improvements obtained in this study.

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